

Noether Sheet 3

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January 11, 2026

These were my solutions to the Noether Mentoring Scheme Sheet 3. Problem statements are abridged.

Problem 1

Suppose a, b, c, d, e and f are successive positive integers. What are the possible values of a if

- (a) $\frac{ef}{a}$ is an integer,
- (b) $\frac{ef}{ab}$ is an integer,
- (c) $\frac{bcd}{e}$ is an integer?

Solution

Write $a = n$ so that $b = (n + 1), c = (n + 2), \dots$ and so on.

(a) We have

$$\frac{ef}{a} = \frac{(n+4)(n+5)}{n} = n + 9 + \frac{20}{n} \in \mathbb{Z}.$$

Thus n must be a factor of 20. The possible values of a are $\{1, 2, 4, 5, 10, 20\}$. $\square$

(b) We have

$$\frac{ef}{ab} = \frac{(n+4)(n+5)}{n(n+1)} = 1 + \frac{8n+20}{n(n+1)} \in \mathbb{Z}.$$

Since $n(n+1) \mid (8n+20)$ we must have $8n+20 \geq n(n+1)$. If one solves the quadratic inequality in n and applies the fact that $n \in \mathbb{N}$ we get that $1 \leq n \leq 9$. Then by testing each value of n we find that the only possible values of a are $\{1, 2, 5\}$. $\square$

(c) We have

$$\begin{aligned} \frac{bcd}{e} &= \frac{(n+1)(n+2)(n+3)}{n+4} \\ &= (n+2)(n+3) - 3(n+4) + 9 - \frac{6}{n+4} \in \mathbb{Z}. \end{aligned}$$

So $(n+4) \mid 6$. Thus $a = n = 2$ is the only possibility. $\blacksquare$

Comments: This problem is just polynomial division, I've omitted the division steps, because I'm lazy.

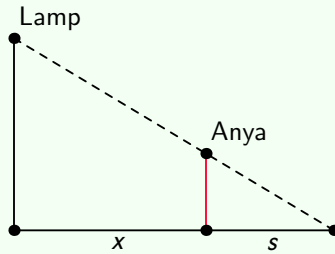
Problem 2

Anya is 2 m tall. She is jogging at 6 m/s away from a 5 m tall streetlight.

How fast is her shadow lengthening?

Solution

Suppose at some point in time Anya is x m away from the streetlight, and her shadow has length s m.



Due to similar triangles we have $x + s = \frac{5}{2}s \implies s = \frac{2}{3}x$. Since x increases by 6 m/s, the length of her shadow increases by $\frac{2}{3}(6) = 4$ m/s. ■

Problem 3

Prove that for every integer $a > 1$, there is a prime p such that $1 + a + a^2 + \cdots + a^{p-1}$ is composite.

Solution

If a is odd then $1 + a$ is an even number greater than 2, which means it must be composite. So for any odd a , we can just pick $p = 2$. So in the rest of the solution, it is assumed we are talking about even $a \geq 2$.

Claim 1: Let $x \in \mathbb{N}$. Modulo 3 the following is true

$$a^x \equiv \begin{cases} 1 & \text{if } x \text{ is even,} \\ a & \text{if } x \text{ is odd.} \end{cases}$$

Proof: Due to Fermat's little theorem we have that $a^2 \equiv 1 \pmod{3}$, (note that $a \neq 3$ because a is even here). Now because of this fact we note that, for positive integer n ,

$$\begin{aligned} a^{2n} &\equiv (a^2)^n \equiv 1 \pmod{3} \\ a^{2n+1} &\equiv a^{2n} \cdot a \equiv a \pmod{3}. \end{aligned}$$

which is the desired result. $\square$

Set $X_p := 1 + a + a^2 + \cdots + a^{p-1}$. In what proceeds take p to be an odd prime. That means we can write $p = 2m + 1$ for some $m \in \mathbb{N}$. Now due to claim 1, reducing mod 3 yields,

$$\begin{aligned} X_p &\equiv 1 + a + 1 + a + \cdots + a + 1 \\ &\equiv \frac{p+1}{2} + \frac{p-1}{2}a \\ &\equiv (m+1) + (m)a. \end{aligned}$$

Now we branch off into three cases. Firstly notice that $X_p \geq 1 + 2 + 2^2 = 7$.

Case 1: $a \equiv 0 \pmod{3}$. Then $X_p \equiv m + 1 \pmod{3}$. Now we can take $m = 2 \implies p = 5$. This gives that $3 \mid X_p$. But since $X_p \geq 7$ it is the case that $X_p \neq 3$ and so X_p is composite.

Case 2: $a \equiv 1 \pmod{3}$. Then $X_p \equiv 2m + 1 \equiv p \pmod{3}$. So simply letting $p = 3$ gives us that $3 \mid X_p$. Moreover since $X_p \geq 7$ we deduce that $X_p \neq 3$ which means that X_p must be composite.

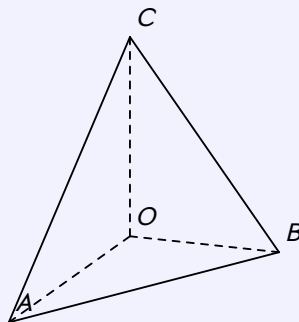
Case 3: $a \equiv 2 \pmod{3}$. Then we can write $a = 3k + 2$ for some $k \in \mathbb{Z}_{\geq 0}$. However now notice that $X_2 = 1 + (3k + 2) = 3(k + 1)$, and if $k \geq 1$ then X_2 is clearly composite. The only remaining case is when $k = 0 \implies a = 2$. To resolve this case notice that when $a = 2$, $X_{11} = 1 + 2 + \cdots + 2^{10} = 2^{11} - 1 = 23 \cdot 89$, which is composite. $\blacksquare$

Comments: The motivation comes from trying to find some sort of telescope-esque trick on X_p modulo something. Once I saw that mod 3 gives an alternating pattern of 1's and a 's, I thought there must be some way to force the expression to be composite.

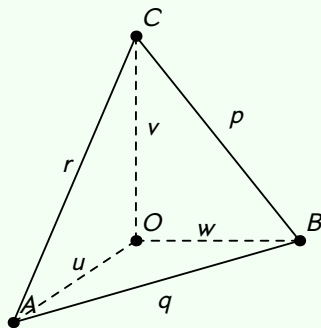
Problem 4

In the tetrahedron with vertices A, B, C and O shown, the angles $\angle AOB, \angle BOC$ and $\angle COA$ are all right angles. Triangles AOB, BOC and COA have areas $6 \text{ cm}^2, \sqrt{39} \text{ cm}^2$ and 5 cm^2 respectively.

What is the area of triangle ABC ?

**Solution**

We label the tetrahedron like so,



Due to the right angle condition we have the following system,

$$\begin{aligned}\frac{1}{2}uw &= 6 \\ \frac{1}{2}vw &= \sqrt{39} \\ \frac{1}{2}uv &= 5.\end{aligned}$$

We solve to find $(u^2, v^2, w^2) = (\frac{20}{13}\alpha, \frac{5}{3}\alpha, \frac{12}{5}\alpha)$ where $\alpha = \sqrt{39}$. We apply Pythagoras,

$$\begin{aligned}q^2 &= u^2 + w^2 = \frac{256}{65}\alpha \\ r^2 &= u^2 + v^2 = \frac{125}{39}\alpha \\ p^2 &= v^2 + w^2 = \frac{61}{15}\alpha.\end{aligned}$$

Now we apply an alternate form of Heron's formula to evaluate $[\triangle ABC]$.

Claim 1: For a triangle with side lengths a, b, c it's area is given by

$$\frac{1}{2} \sqrt{(bc)^2 - \left(\frac{b^2 + c^2 - a^2}{2} \right)^2}$$

Proof: Refer to Noether Sheet 2, problem 4. $\square$

Let $q = a, r = b, p = c$ then $(bc)^2 = \frac{1525}{117} \alpha^2$. Also

$$\frac{b^2 + c^2 - a^2}{2} = \frac{1}{2} \frac{10}{3} \alpha = \frac{5}{3} \alpha.$$

Now using the formula

$$[\triangle ABC] = \frac{1}{2} \sqrt{\frac{1525}{117} \alpha^2 - \frac{25}{9} \alpha^2} = \frac{1}{2} \alpha \frac{20}{\sqrt{39}} = 10. \blacksquare$$

Problem 5

Let p be a prime and let a be an integer coprime to p .

- (a) Show that $ax \equiv ay \pmod{p}$ implies that $x \equiv y \pmod{p}$.
- (b) What can you say about the set of numbers $\{a, 2a, 3a, \dots, (p-1)a\}$ when working modulo p .
- (c) What is the value of $a^{p-1} \pmod{p}$?

Solution

(a) By definition $a(x - y)$ is a multiple of p . But since $\gcd(a, p) = 1$ it must be the case that $p \mid (x - y)$ which is equivalent to the desired result. $\square$

(b) Now due to part (a) none of the numbers in that set are congruent mod p . So it is the case that when working mod p ,

$$\{a, 2a, 3a, \dots, (p-1)a\} = \{1, 2, 3, \dots, (p-1)\}. \quad \square$$

(c) Because those two sets are the same mod p , their products are the same mod p also,

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Now due to (a) this implies that $a^{p-1} \equiv 1 \pmod{p}$. $\blacksquare$

Problem 6

Suppose that ABC is a triangle with circumradius R . Let the lines BC, CA and AB have lengths a, b, c respectively.

Prove that

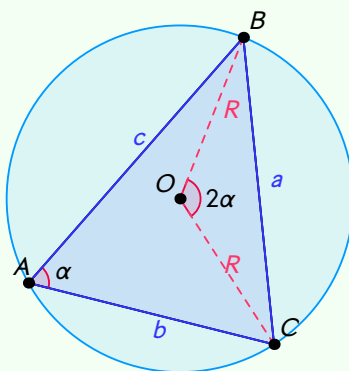
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

where A, B and C refer to the three angles of the triangle ABC .

This is the *extended sine rule*.

Solution

Call $\angle A = \alpha$. It is a well-known fact that $\angle BOC = 2\angle BAC = 2\alpha$.



Now by applying the cosine rule on $\triangle BOC$, followed by the double angle formula yields

$$\begin{aligned} a^2 &= 2R^2 - 2R^2 \cos 2\alpha \\ &= 2R^2(1 - (-1 + 2\cos^2 \alpha)) \\ &= 4R^2(1 - \cos^2 \alpha) \\ &= 4R^2 \sin^2 \alpha. \end{aligned}$$

Notice that since $0^\circ < \alpha < 180^\circ$, $\sin \alpha > 0$ and $a > 0$ by definition. So we deduce that

$$\frac{a}{\sin \alpha} = 2R.$$

Then by a symmetrical argument (or just by the normal sine rule) we get that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R,$$

as desired. ■

Problem 7

Find the roots of $x^2 - 10^{10}x + 1$, accurate to 30 decimal places.

Solution

By the quadratic formula the solutions look like

$$x = \frac{10^{10}}{2} \pm \frac{1}{2} \left(10^{20} - 4 \right)^{1/2}$$

We just need to approximate the second term to 30 decimal places, since the first term is an integer. In what follows let $(x)_n$ denote the falling factorial. That is $(x)_n = x(x-1) \cdots (x-n+1)$, $n \geq 1$ and $(x)_0 = 1$. Define $\alpha := 10^{20}$. By the generalised binomial theorem,

$$\begin{aligned} \frac{1}{2} (\alpha - 4)^{1/2} &= \frac{1}{2} \sum_{r=0}^{\infty} \frac{(1/2)_r}{r!} \alpha^{1/2-r} (-4)^r \\ &= \frac{\sqrt{\alpha}}{2} \sum_{r=0}^{\infty} \frac{(1/2)_r}{r!} \left(\frac{-4}{\alpha} \right)^r \\ &> \frac{\sqrt{\alpha}}{2} \left(\frac{1}{1} (1) + \frac{1/2}{1} \left(\frac{-4}{\alpha} \right) + \frac{(1/2)(-1/2)}{2} \left(\frac{-4}{\alpha} \right)^2 \right) \end{aligned}$$

This approximation is sufficient since every other term will have magnitude less than roughly $\sim 1/\alpha^{2.5} = 10^{-50}$. We compute our approximation,

$$\frac{10^{10}}{2} - \frac{1}{10^{10}} - \frac{1}{10^{30}}.$$

So now putting that together the two roots (to 30 decimal places) are

$$x = 10^{10} - \frac{1}{10^{10}} - \frac{1}{10^{30}} \text{ and } \frac{1}{10^{10}} + \frac{1}{10^{30}}. \blacksquare$$

Problem 8

17:36 29/05/84 is a *pandigital* time and date as it uses each of the digits 0-9 exactly once.

The *product* of a date and time is the product of the five two-digit numbers that make it up. For example, the product of the pandigital example above is $17 \times 36 \times 29 \times 05 \times 84$.

Find all pandigital dates and times for which the product is a perfect cube.

Solution

I am of the opinion that problems like number 8 are far more amenable to informatics competitions, since the problem looks to be a very long case bashing problem. So instead of a mathematical solution I present one such programatic solution in Python, after all computer science is all just applied math!

```

1  # Noether Sheet 3 - Problem 8
2  from itertools import permutations
3
4  def Validate_Date_Time(h, minn, d, mon, y):
5      if not 1 <= h <= 23: return False
6      if not 1 <= minn <= 59: return False
7      if not 1 <= d <= 31: return False
8      if not 1 <= mon <= 12: return False
9
10     if mon == 2:
11         if d > 28: return False
12
13     if mon in [4, 6, 9, 11]:
14         if d > 30: return False
15
16     return True
17
18  def is_perfect_cube(n):
19      lo, hi = 0, int(n ** (1/3)) + 2
20      while lo <= hi:
21          mid = (lo + hi) // 2
22          cube = mid ** 3
23
24          if cube == n: return True
25          elif cube < n: lo = mid + 1
26          else: hi = mid - 1
27
28      return False
29
30  NUMBERS = "0123456789"
31  for p in permutations(NUMBERS, 10):
32      hour = int(p[0] + p[1])
33      minute = int(p[2] + p[3])
34      day = int(p[4] + p[5])
35      month = int(p[6] + p[7])
36      year = int(p[8] + p[9])

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37
38     if Validate_Date_Time(hour, minute, day, month, year):
39         product = hour * minute * day * month * year
40
41         if is_perfect_cube(product):
42             print(f"{hour}:{minute} {day}/{month}/{year}")
```

After running the code, it takes around 5 seconds to run, the output is 13:54 26/9/78 only. ■